MILITARY SPECIFICATION

RADIOACTIVE TEST SAMPLE, KRYPTON 85,

MX7338/PDR-27R

1. SCOPE

1.1 This specification covers one type of radioactive source encased in a copper capsule then sealed in a small aluminum rod.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated Plywood.
PPP-B-676	-	Boxes, Set-Up, Paperboard.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet Stock, (Container Grade) and Cut Shapes.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-M-19590	-	Marking of Commodities and Containers to Indicate Radioactive Material.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

FSC 6665

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DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL124-12-6 - Radioactive Test Sample, Krypton 85, MX7338/PDR-27R.
D150-2-1 - Marking Diagram for Shipping Containers.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Officer, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL124-12-6 or on subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawing.

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3.1.2 Components. All components cited on Drawing DL124-12-6 or on subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawing.

3.1.2.1 Radiation source. The source of radiation shall consist of 5 millicuries ± 10 percent of Krypton 85, hermetically sealed in a copper capsule as specified on Drawing B124-12-8.

3.1.2.2 Hermetic seal. The capsule shall show no evidence of air leakage when submerged in a constant temperature bath as specified in 4.4.4.1.

3.2 Manufacture and assembly. The radioactive test sample shall be assembled in accordance with Drawing C124-12-6.

3.3 Activity. The radioactive test sample shall have an activity of 5 millicuries ± 10 percent when tested as specified in 4.4.4.2.

3.4 Mechanical shock. The activity of the radioactive test sample shall not vary more than ± 5 percent from the initial activity after being subjected to the drop test as specified in 4.4.4.3.

3.5 Thermal shock. The activity of the radioactive test sample shall not vary more than ± 5 percent from the initial activity after being subjected to three cycles of thermal shock as specified in 4.3.3.1.

3.6 Preproduction. Prior to the start of regular production, preproduction samples of the radioactive test sample shall be produced in accordance with this specification for examination and test (see 4.3.)

3.7 Workmanship. The radioactive test sample shall be free from cracks, dents, burrs, abrasions, faulty seals, and foreign matter (dirt, oil, and viscous material.)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

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4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 50 radioactive test samples shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. Each radioactive test sample shall be examined and tested for all requirements of the applicable drawings and this specification. Tests shall be performed in accordance with the number and sequence in Table I.

Table I. Preproduction Test Sequence

Item	Drawing	Sample size	Test
Capsule, Krypton 85	B124-12-8	50	Hermetic seal 4.4.4.1
Assembled Test Source	C124-12-6	50	Activity 4.4.4.2
Assembled Test Source	C124-12-6	25	Mechnical Shock 4.4.4.3
Assembled Test Source	C124-12-6	25	Thermal Shock 4.3.3.1

4.3.3 Tests.

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4.3.3.1 Thermal shock. The 25 radioactive test samples from the 50 preproduction samples shall be subjected to a temperature of -40°C + 2°C for 2 hours. At the end of the 2 hour period the test samples shall be subjected to a temperature of +50°C + 2°C for 2 hours. This procedure shall constitute one complete cycle of the thermal test. Three continuous cycles shall be performed. Upon completing the last cycle, the activity shall be determined in accordance with 4.4.4.2.

4.3.4 Acceptance/rejection criteria. The acceptance number for all tests shall be zero.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the radioactive test samples (Drawing C124-12-6) produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For mechanical shock. Sampling shall be conducted in accordance with MIL-STD-105, level S-3 and an AQL of 1.0 percent.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. The sample items and the level A preparation for delivery shall be examined and tested in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 For mechanical shock. Sample items shall be tested in accordance with 4.4.4.3 and with MIL-STD-105, level S-3 using an AQL of 1.0 percent defective for acceptance.

4.4.3.3 For critical defects. Each item in the lot shall be inspected for critical characteristics in the classification of defects.

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4.4.3.4 Classification of defects.

(a) Radioactive test sample, Drawing C124-12-6.

Categories	Defects	Acceptance standard
Critical:		
1	Activity	4.4.4.2
2	Color missing from radioactive end	
3	Marking missing, illegible, incorrect or incomplete	
Major:	AQL 1.0 percent defective	
101	Body dimensions	
102	Body not properly sealed	
103	Chain missing	
104	Incorrect chain length	
105	Tag not firmly attached to sample	
106	Workmanship (see 3.7)	

(b) Capsule, Drawing B124-12-8.

Categories	Defects	Acceptance standard
Critical:		
1	Hermetic seal	4.4.4.1
Major:		
101	Length 1.203	

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4.4.3.4 Classification of defects (continued)

(c) Preparation for delivery (section 5).

Categories	Defects
Critical:	
1	Marking of unit or shipping container illegible, incorrect, incomplete or not durable
Major:	AQL 1.0 percent defective
101	Unit or shipping container not as specified
102	Quantity of samples per unit or shipping container not as specified or indicated
103	Closure of unit or shipping container not as specified
104	Unit or shipping container damaged
105	Cushioning of shipping container not as specified

4.4.4 Tests.

4.4.4.1 Hermetic seal. Submerge each capsule (Drawing B124-12-8) for a minimum of 60 seconds in a suitable constant temperature bath such as glycerine heated to 150 ± 5°F. A steady stream or recurrent sucession of bubbles from the end of the capsule shall indicate leakage.

4.4.4.2 Activity. The radioactive test sample (Drawing C124-12-6) shall be tested for activity using a calibrated scintillation or Geiger-Mueller probe connected to a scaler or spectrometer. Each radioactive test sample shall be numbered and the activity level recorded (initial reading).

4.4.4.3 Mechanical shock. The radioactive test sample (Drawing C124-12-6) shall be tested by dropping from a height of 4 feet and impacting on a steel surface. Each radioactive test sample shall be dropped six times then subjected to the activity test 4.4.4.2 and the results recorded.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. Radioactive test samples shall be individually packaged, method IC-3 of MIL-P-116, in a paperboard box, of optional type and class, conforming to PPP-B-676. The test sample shall be sufficiently cushioned to fill all voids with material conforming to PPP-C-843.

5.1.2 Level C. The radioactive test samples shall be packaged to afford adequate protection against deterioration and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing.

5.2.1 Level A. Radioactive test samples, packaged as specified in 5.1.1, shall be packed in a quantity not to exceed the applicable requirement specified in Department of Transportation (DOT) regulations 49CFR 171-179 in a cleated plywood box conforming to overseas type, style J of PPP-B-601, for a type 2 average load. Unless otherwise specified, the plywood shall be provided with water repellent preservative treatment as specified in PPP-B-601, in addition, wood cleats shall be immersed for a minimum of three minutes in the same preservative as that specified for the plywood (see 6.2). Each inside face of the wood box shall be lined with fiberboard sheets conforming to grade W5c of PPP-F-320. Additional sheets shall be added as required to assure a tight pack. Boxes shall be closed and strapped using galvanized strapping as specified in the appendix to PPP-B-601.

5.2.2 Level C. Radioactive test samples, packaged as specified in 5.1.2, shall be packed in accordance with applicable requirements specified in DOT regulation 49 CFR 171-179 to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall comply with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with MIL-STD-129, DOT regulation 49 CFR 171-179 and MIL-M-19590. Shipping containers shall be marked as shown on Drawing D150-2-1.

6. NOTES

6.1 Intended use. The radioactive test samples covered by this specification are intended to provide a radiation source that permits the operator to ascertain the operating condition of AN/PDR 27 radiac set when no known radiation field is available.

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6.2 Ordering data. Procurement documents should specify the following:

(a) The title, number, and date of this specification.
(b) Level of packaging and packing required.
(c) Quantity required in each shipping container.
(d) When wood preservative is not required on shipping container.

6.3 Caution. Care should be exercised to prevent injury to personnel engaged in handling and testing radioactive sources. The National Bureau of Standards handbook (NBS No. 73) and Atomic Energy Commission Regulation 10 CFR 20-40 contain information pertaining to protection of personnel from radioactive emanations.

Custodian:

Army - MU

Preparing activity:

Army - MU (EA)

Project No. 6665-A217

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel of Government activities involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 [illegible] WHICH MAY BE USED.

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